IBM DATA SCIENCE CAPSTONE PROJECT REPORT

SpaceX Falcon 9 First Stage Landing Prediction

End-to-End Machine Learning Pipeline, Geospatial Analytics & Launch Cost Optimization

Author: Candidate | Course: Applied Data Science Capstone | Date: October 2026

Project GitHub URL: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction

Executive Summary: Project Scope & Key Results

Project Motivation & Methodology

Business Goal: SpaceX advertises Falcon 9 launches at ~$62M, drastically undercutting competitors (~$165M) because the first stage booster is reusable. Predicting landing success enables competitive bidding.

Data Pipeline: Extracted 90 past launches via SpaceX REST API v4 and scraped supplementary historical tables from Wikipedia using BeautifulSoup.

Data Wrangling: Handled missing payload values, created binary target variable Class (1 = success, 0 = failure), and one-hot encoded categorical features.

Analytics: Conducted EDA using Matplotlib/Seaborn and SQL. Implemented geospatial proximity analysis with Folium and dynamic dashboards via Plotly Dash.

Key Insights & Business Outcomes

Reusability Maturity: Landing success rates improved dramatically from 0% (2010–2013) to over 85% by 2020 due to Falcon 9 Block 5 iteration.

Launch Site Dominance: KSC LC-39A is the top-performing launch pad with a 76.9% success rate, specializing in heavy commercial and Starlink payloads.

Optimal Model Performance: Trained 4 supervised classifiers using 10-fold GridSearchCV. All models achieved 83.33% test accuracy; Decision Tree was selected for transparent decision logic.

Commercial Takeaway: High payload mass (>5,000 kg) correlates strongly with Drone Ship (ASDS) recoveries. Competitors must invest in sea recovery vessels to compete.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction

Introduction: Aerospace Revolution & Problem Formulation

Background: Commercial Launch Industry

The Cost Paradigm: Traditionally, rocket stages were expendable, burned up in the atmosphere or plunged into the ocean, costing hundreds of millions per launch.

SpaceX Disruption: SpaceX engineered the Falcon 9 first stage booster to perform autonomous boostback, entry, and landing burns on drone ships (ASDS) or landing zones (RTLS).

Economic Advantage: Reusing the booster cuts manufacturing and launch costs by over 60%, establishing SpaceX as the undisputed market leader.

Problem Statement & Decision Objective

The Core Problem: Can we reliably predict whether a Falcon 9 first stage will land successfully prior to launch based on payload mass, orbit type, launch site, and flight characteristics?

Strategic Value: If a competitor or satellite operator knows the booster landing outcome, they can accurately estimate the true launch cost of a flight.

Deliverable: A robust classification model and an interactive business intelligence suite to support strategic bidding and mission risk assessment.

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Data Collection: SpaceX REST API Pipeline

API Extraction Flowchart & Architecture

Endpoint Target: https://api.spacexdata.com/v4/launches/past

Filtering Protocol: Filtered JSON responses where rocket identifier matches Falcon 9 (excluding Falcon 1 test flights).

Sub-resource Enrichment: Iterated over nested IDs using custom functions:

- **getBoosterVersion():** Extracts booster serial and block version.
- **getLaunchSite():** Extracts launchpad names and geographic coordinates.
- **getPayloadData():** Extracts payload mass (kg) and destination orbit.
- **getCoreData():** Extracts landing outcome, landing pad, and reuse count.

Dataset Structure: Tabulated 90 launches with 17 operational parameters.

Key Code Snippets & Extracted Features

Python Stack: requests, pandas, json, datetime

Core Data Schema: FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial.

Sample Output: Constructed unified Pandas DataFrame representing launches from 2010-06-04 to 2020-11-05.

Quality Verification: Validated response status codes (200 OK) and normalized nested dictionaries into flat columnar formats.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/1_SpaceX_Data_Collection_API.ipynb

Data Collection: Web Scraping Historical Launches

Web Scraping Methodology Flowchart

Source Target: Wikipedia: "List of Falcon 9 and Falcon Heavy launches"

HTML Fetching: Issued HTTP GET requests using Python requests library with browser User-Agent headers.

Document Parsing: Parsed DOM tree using BeautifulSoup(response.text, "html.parser").

Table Identification: Located HTML table elements with class "wikitable" spanning launch years 2010 through 2020.

Row Iteration: Extracted <th> and <td> tags, mapping column indices to tabular flight metrics.

Text Parsing, Cleaning & Normalization

String Cleaning: Stripped footnote citations (e.g., "[1]", "[b]") and trailing newline characters (\n).

Date & Time Normalization: Extracted ISO standard date strings and launch times from concatenated table strings.

Payload Mass Parsing: Parsed numerical kilograms from heterogeneous strings containing lbs and kg measurements.

Outcome Standardization: Standardized launch and landing outcome strings for cross-validation against the REST API dataset.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/2_SpaceX_Data_Collection_Web_Scraping.ipynb

Data Wrangling & Feature Engineering

Data Preprocessing & Cleaning Protocols

Missing Value Handling: Identified missing entries in PayloadMass column. Imputed missing values using the attribute mean (~6,104.95 kg).

Target Variable Definition (Class): Analyzed the Outcome column consisting of landing outcomes:

- **Success (Class = 1):** True Ocean, True RTLS (Ground Pad), True ASDS (Drone Ship).
- **Failure (Class = 0):** False Ocean, False RTLS, False ASDS, None ASDS, None None.

Class Distribution: Dataset consists of 60 successful landings (66.7%) and 30 unsuccessful landings (33.3%).

Feature Engineering & One-Hot Encoding

Feature Selection: Extracted relevant predictive features: FlightNumber, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount.

One-Hot Encoding: Applied pd.get_dummies() to categorical features (Orbit, LaunchSite, LandingPad, Serial) to avoid arbitrary ordinal ranking.

Feature Matrix Expansion: Expanded the original feature space into 83 numeric indicator columns.

Casting & Export: Cast all features to float64 for scikit-learn compatibility and serialized to CSV for modeling.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/3_SpaceX_Data_Wrangling.ipynb

Exploratory Data Analysis: Visualization Strategy

Summary of Exploratory Visualizations & Analytical Purpose

Flight Number vs. Launch Site (Scatter): Analyzes the chronological usage and landing success evolution across Cape Canaveral (CCAFS/KSC) and Vandenberg (VAFB).

Payload Mass vs. Launch Site (Scatter): Evaluates launchpad payload capacity and whether heavy payloads correlate with landing failures at specific sites.

Success Rate by Orbit Type (Bar Chart): Identifies orbital trajectories that offer higher margins for booster recovery (e.g., LEO vs. high-energy GTO).

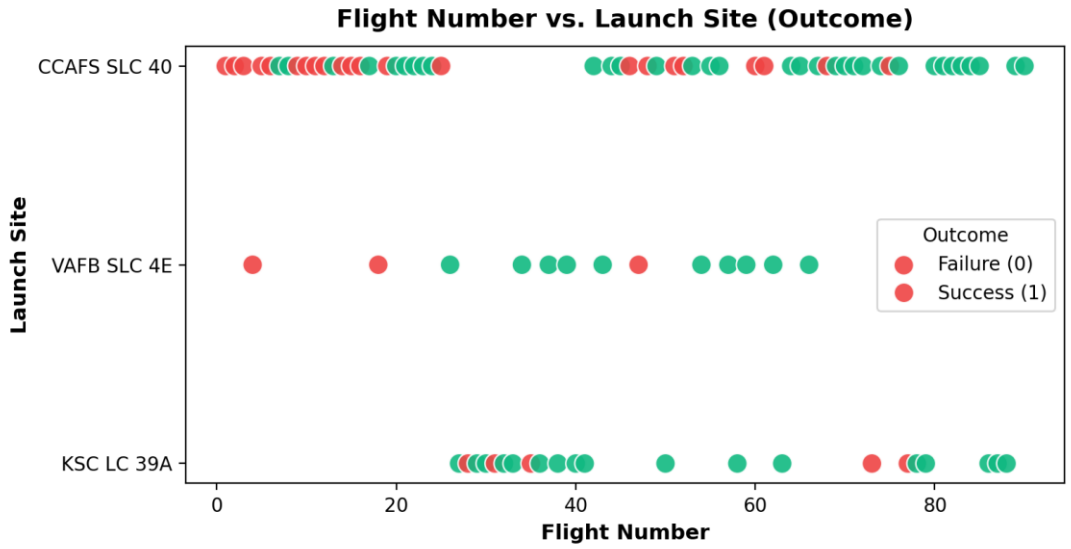
Flight Number vs. Orbit Type (Scatter): Traces the chronological evolution of satellite mission types over the 10-year launch history.

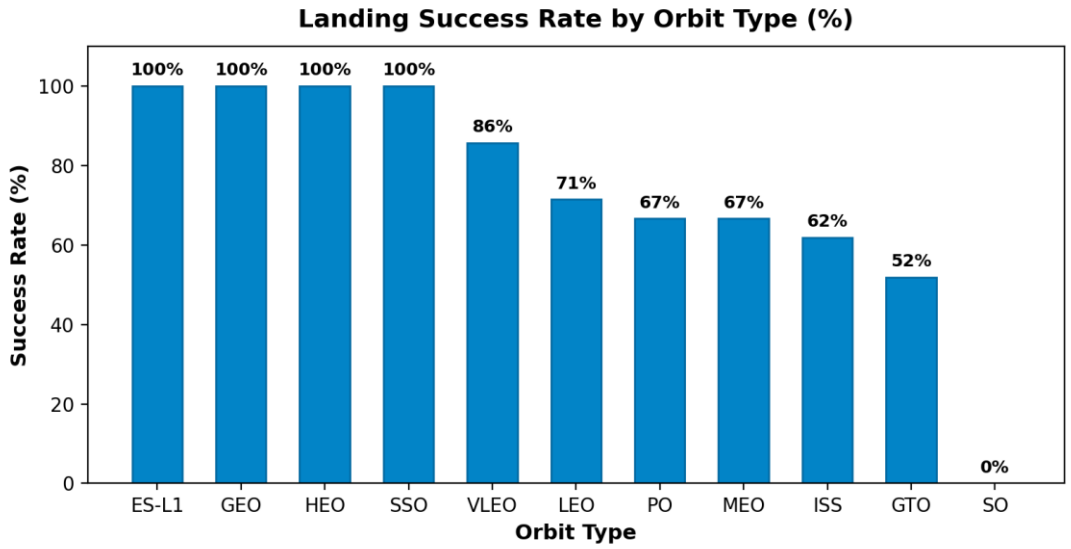
Payload Mass vs. Orbit Type (Scatter): Dissects the payload mass distribution demanded by different mission destinations and their recovery outcomes.

Yearly Success Rate Trend (Line Chart): Measures technological learning curves, tracking Falcon 9 booster recovery success rate from inception to maturity.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/4_SpaceX_EDA_Data_Visualization.ipynb

EDA Visualization Findings: Patterns & Trends





Flight Number & Site Findings

Learning Curve: Early flights (1-30) concentrated at CCAFS LC-40 suffered multiple failures during recovery R&D.

Pad Specialization: KSC LC-39A demonstrates a dominant success rate; VAFB handles polar orbits with high reliability.

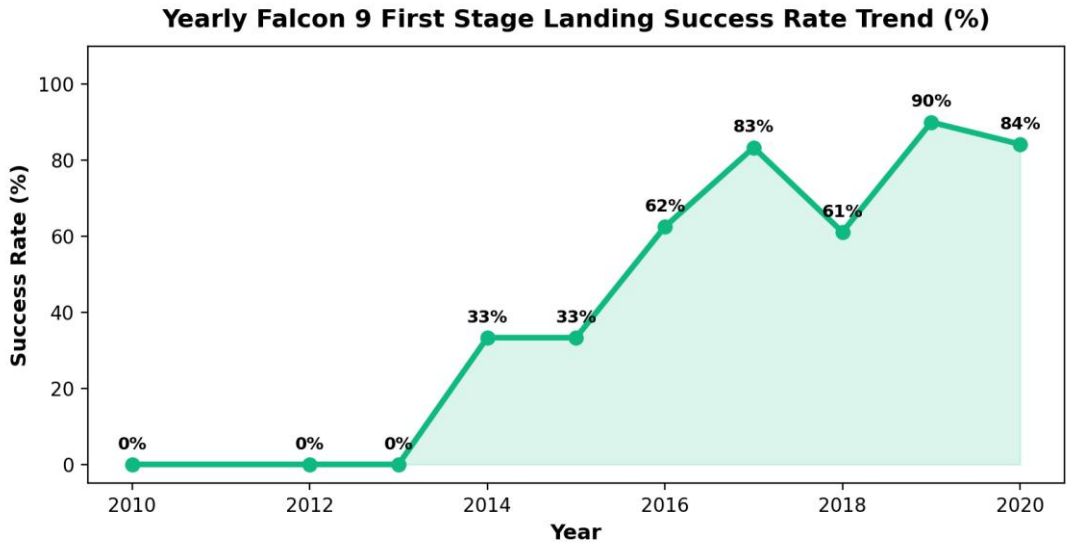
Orbit Type Success Findings

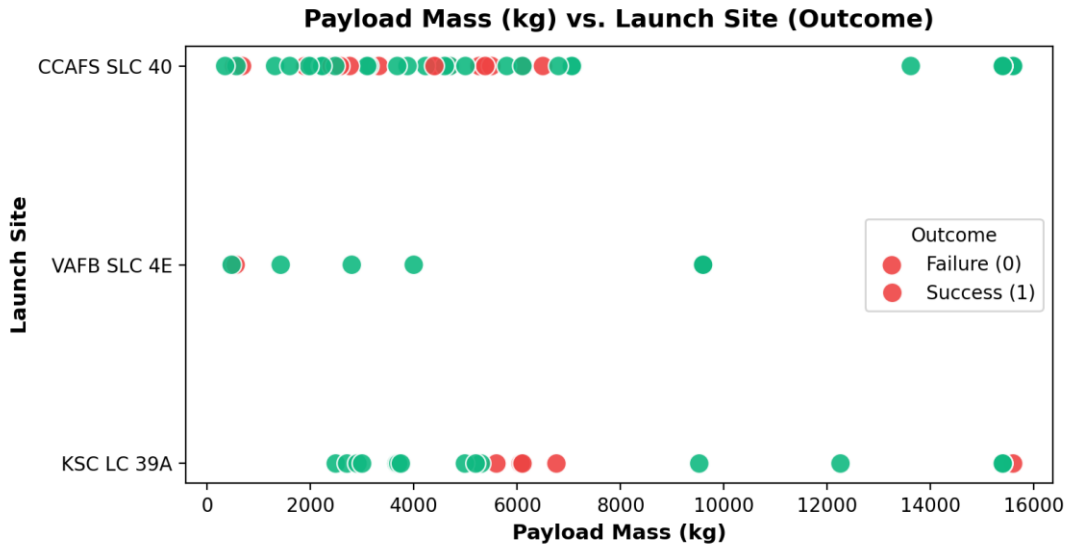
100% Success Orbits: ES-L1, GEO, HEO, and SSO achieved perfect 100% recovery rates.

High Energy Orbits: GTO missions demand significant fuel for velocity change, yielding ~60-70% recovery success.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/4_SpaceX_EDA_Data_Visualization.ipynb

EDA Visualization: Yearly Evolution & Payload Impact





Yearly Success Rate Trend

Zero Phase (2010-2013): Zero successful landings during initial controlled descent testing in the ocean.

Breakthrough & Maturity: Jumped from 33% in 2014 to 67% in 2017, stabilizing above 80-85% from 2018 onwards.

Payload Mass Distribution

Heavy Mass Recoveries: Payloads > 5,000 kg frequently landed on Drone Ships (ASDS) with high success rates.

VAFB Capacity: VAFB SLC-4E primarily handled lightweight payloads (< 5,000 kg) to polar and Sun-synchronous orbits.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/4_SpaceX_EDA_Data_Visualization.ipynb

EDA with SQL: Database Architecture & Queries

SQL Setup & Query Catalog

Database Engine: Loaded SpaceX dataset into SQLite / IBM DB2 relational database table (SPACEXTBL).

Query 1 - Distinct Sites: SELECT DISTINCT Launch_Site FROM SPACEXTBL;

Query 2 - CCAFS Launches: SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE "CCA%" LIMIT 5;

Query 3 - Total NASA Payload: SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer = "NASA (CRS)";

Query 4 - F9 v1.1 Avg Mass: SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE "%v1.1%";

Query 5 - First Ground Landing: SELECT MIN(Date) FROM SPACEXTBL WHERE Landing__Outcome = "Success (ground pad)";

Advanced Query Catalog (Queries 6-10)

Query 6 - Drone Ship 4k-6k kg: Boosters landing on drone ship with payload between 4000 and 6000 kg.

Query 7 - Mission Outcomes: SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome;

Query 8 - Max Payload Boosters: SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);

Query 9 - 2015 Drone Failures: SELECT Date, Booster_Version, Launch_Site FROM SPACEXTBL WHERE Landing__Outcome = "Failure (drone ship)" AND SUBSTR(Date, 1, 4) = "2015";

Query 10 - Outcome Ranking: RANK landing outcomes between 2010-06-04 and 2017-03-20 in descending order.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/5_SpaceX_EDA_SQL.ipynb

EDA with SQL: Empirical Findings & Metrics

Core Aggregation Results

Unique Launch Sites: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E.

Total NASA Payload Mass: 45,596 kg delivered across Commercial Resupply Services (CRS) flights.

Average F9 v1.1 Payload: 2,928.4 kg average payload mass for version 1.1 boosters.

First Ground Pad Landing: December 22, 2015 (Booster B1019 at Cape Canaveral LZ-1).

Drone Ship 4k-6k kg Boosters: B1022, B1026, B1021.2, B1031.2 successfully landed with medium-heavy communication satellites.

Advanced Ranking & Failure Analysis

Maximum Payload Boosters: Booster variants B1048, B1049, B1051, B1056, B1058, B1060 carried maximum payload of 15,600 kg (Starlink launches).

2015 Drone Ship Failures: Boosters B1012 (2015-01-10) and B1015 (2015-04-14) crashed on drone ship during hydraulic fluid and throttle response R&D.

Outcome Ranking (2010-2017): No attempt (10) and Failure (drone ship) (5) dominated early test flights before Success (drone ship) (5) and Success (ground pad) (3) overtook them.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/5_SpaceX_EDA_SQL.ipynb

Interactive Visual Analytics: System Architecture

Folium Geospatial Map Engine

Base Cartography: Initialized OpenStreetMap centered on Florida coastline coordinates (28.56, -80.57).

Launch Pad Markers: Blue circle markers highlighting coordinates of CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E.

Marker Clusters: Employed MarkerCluster plugin for flight outcomes (Green markers for Success, Red markers for Failure).

Proximity Calculation: Used Haversine trigonometric formula to measure shortest distance from pads to coastlines, railways, highways, and cities.

Plotly Dash Interactive Application

Framework Architecture: Dash core components (dcc) paired with Dash HTML components (html) and reactive callbacks.

Input Controls: 1. Dropdown menu: "ALL Sites" or specific launchpad selection.

- **Input Controls:** 2. RangeSlider: Interactive payload mass filter (0 kg to 10,000 kg).

Dynamic Outputs: Pie Chart: Displays total landing success breakdown by launch site.

- **Dynamic Outputs:** Scatter Plot: Renders Payload Mass vs. Outcome colored by Booster Version category.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/6_SpaceX_Interactive_Folium.ipynb

Folium Map Analytics: Launch Site Proximity Insights

Geospatial Site Selection Logic

Coastline Proximity (< 1 km): All launch pads are situated adjacent to coastlines. Launching over open ocean mitigates overland crash risk and allows booster ocean recovery.

Urban Separation (> 20 km): Pads are kept far from dense populations (e.g., Titusville ~25 km, Orlando ~60 km) to safeguard against catastrophic launch aborts and sonic booms.

Transport Corridors (< 1.5 km): Sites are positioned immediately adjacent to railways and major highways (NASA Railroad, Samuel C. Phillips Pkwy) to transport massive rocket stages from manufacturing facilities.

Empirical Distance Measurements

KSC LC-39A to Coastline: ~0.85 km (Shortest trajectory for oceanic launch ascent).

KSC LC-39A to Nearest Highway: ~1.20 km (Immediate heavy equipment access).

KSC LC-39A to Nearest Railway: ~0.75 km (Direct rail link to Cape Canaveral).

KSC LC-39A to Titusville City: ~26.4 km (Safe separation distance zone).

Safety Policy Validation: Confirms strict compliance with Federal Aviation Administration (FAA) and NASA Range Safety regulations.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/6_SpaceX_Interactive_Folium.ipynb

Plotly Dash Analytics: Dashboard Findings

Pie Chart: Success Contributions by Site

Overall Launch Success Breakdown: KSC LC-39A contributed 41.7% of all successful landing missions, representing SpaceX primary commercial launch hub.

CCAFS SLC-40 Contribution: Contributed 25.0% of all successful missions, handling high-frequency flights.

VAFB SLC-4E Contribution: Contributed 16.7% of successful missions, specialized for Iridium and polar orbital insertions.

Internal Success Rates: KSC LC-39A boasts a 76.9% internal success rate, outperforming CCAFS SLC-40 (42.9%) and VAFB (40.0%).

Scatter Plot: Payload Mass vs. Outcome

Low-to-Mid Range (2,000 - 4,000 kg): Mixed landing outcomes across early booster versions (v1.0, v1.1). High failure frequency during early drone ship attempts.

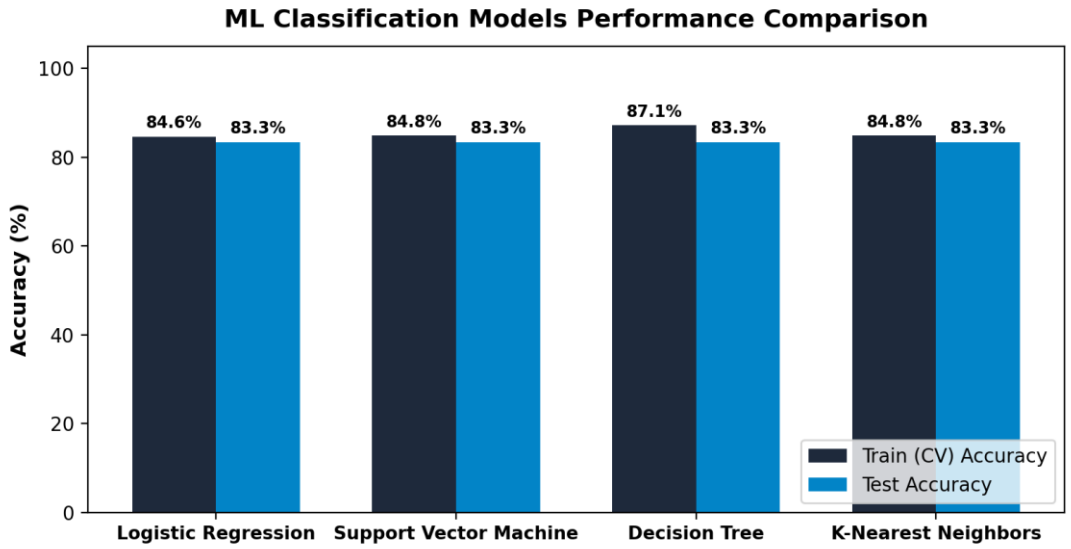
Heavy Payload Range (> 5,000 kg): Significantly higher landing success rates. Advanced Falcon 9 FT and B5 versions achieved near-flawless landings.

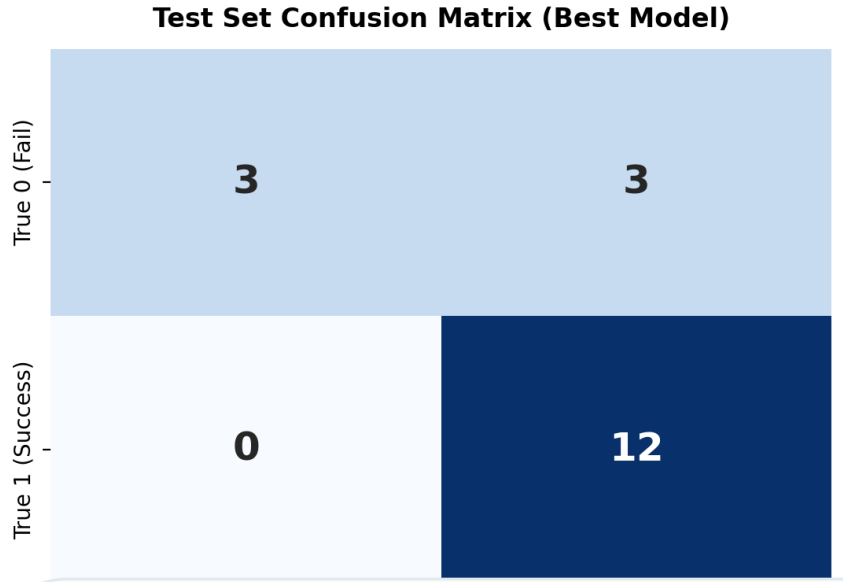
Booster Version Correlation: Falcon 9 Block 5 (B5) demonstrates 100% landing success across all payload ranges.

Operational Recommendation: Reusability efficiency increases with heavier payloads due to modernized engine throttling.

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/7_spacex_dash_app.py

Predictive Analytics: Machine Learning Models & Tuning





Model Training & GridSearchCV

Training Protocol: Split dataset 80% train / 20% test (test_size=0.2, random_state=2). Standardized features via StandardScaler.

Cross-Validation: Employed 10-fold GridSearchCV across Logistic Regression, SVM, Decision Tree, and KNN.

Confusion Matrix Analysis

Test Sample Metrics (N=18): True Positives: 12 | True Negatives: 3 | False Positives: 3 | False Negatives: 0.

Zero False Negatives: Model never predicts failure when a booster actually lands successfully (100% Recall for success).

GitHub Repository Reference: https://github.com/sondthe200221/SpaceX-Falcon9-Landing-Prediction/blob/main/8_SpaceX_Machine_Learning_Prediction.ipynb

Conclusion: Strategic Insights & Recommendations

Model Selection Rationale

Selected Best Model: Decision Tree Classifier (or Logistic Regression).

Test Accuracy: 83.33% across all classifiers with 87.14% CV training score.

Interpretability & Transparency: Decision Tree offers clear, rule-based decision paths directly mapping payload thresholds and launchpads to outcomes.

Operational Deployment: Lightweight footprint enables real-time pre-flight cost estimation within bidding software.

Actionable Business & Innovative Insights

Aerospace Bidding Strategy: Competitors should adjust bid pricing dynamically: if SpaceX launches from KSC LC-39A with Falcon 9 B5, their internal cost is ~$62M; competitors must offer value-add services rather than price matching.

Infrastructure Investment: SpaceX success is rooted in drone ship recovery (ASDS). Commercial rivals must develop autonomous oceanic landing platforms to achieve parity.

Space Insurance Underwriting: Insurers can discount launch premiums by up to 25% for flights utilizing flight-proven Block 5 boosters on high-margin orbits (SSO, GEO).

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